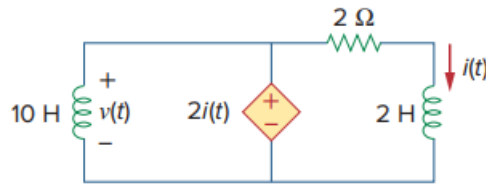


Problem

For the circuit in Fig. 8.101, find $v(t)$ for $t > 0$. Assume that $i(0^+) = 2$ A.

Figure 8.101:



Step-by-step solution

Step 1 of 3

Refer to the Figure 8.55 in the textbook.

Redraw the circuit as shown in Figure 1.

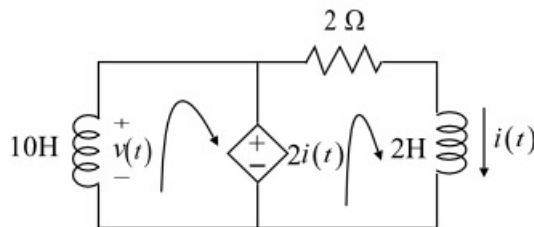


Figure 1

Step 2 of 3

Apply Kirchhoff's voltage law around the first loop.

$$-v(t) + 2i(t) = 0$$

$$v(t) = 2i(t) \dots\dots (1)$$

Apply Kirchhoff's voltage law in the second loop.

$$-2i(t) + 2i(t) + 2 \frac{di(t)}{dt} = 0$$

$$\frac{di(t)}{dt} = 0$$

Integrate both the sides with respect to t .

$$\int \frac{di(t)}{dt} dt = \int 0 dt$$

$$i(t) = k \dots\dots (2)$$

Here, k is a constant.

Step 3 of 3

Substitute 0 for t , 2 A for $i(0)$ in equation (2).

$$i(0) = k$$

Therefore, the value of $i(t)$ is 2 A.

$$k = 2$$

Substitute 2 for $i(t)$ in equation (1).

$$\begin{aligned} v(t) &= 2(2) \\ &= 4 \text{ V} \\ &= 4u(t) \text{ V} \end{aligned}$$

Therefore, the voltage $v(t)$ across the 10 H inductor is $\boxed{4u(t) \text{ V}}$.